

Final Report for Database Management

Project (UCS-310)

on

MISSING PERSONS & RE-UNION TRACKING SYSTEM

B-Tech (2nd Year)

Batch/Group: 2Q14

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MISSING PERSONS & RE-UNION TRACKING SYSTEM

Application Domain

The application domain of this project lies in the field of public safety and social welfare. The system is intended to support police departments, NGOs, shelters, orphanages, and rehabilitation centres in managing information related to missing persons and reunification processes. It provides a centralized platform for maintaining records of missing individuals, unidentified persons, sponsorships, shelters, and case progress.

The project also falls under the domain of database-driven management systems, where multiple organizations interact and share information through a common relational database. The system enables efficient coordination between different stakeholders involved in rescue, rehabilitation, and sponsorship activities while ensuring proper data organization, integrity, and accessibility.

Motivation for Selecting the Problem

The motivation behind this project arises from the increasing number of missing person cases and the lack of a centralized system to manage and track such information efficiently. Existing systems often rely on manual records or isolated databases, which can lead to duplication of information, poor communication, and inefficient case handling.

This project aims to provide a structured and centralized database solution that improves transparency, coordination, and accessibility of information among different organizations. By integrating case management, shelter tracking, sponsorship management, and audit logging into a single platform, the system helps streamline the overall process of search, rehabilitation, and reunification. The project also demonstrates the practical application of DBMS concepts such as normalization, relationships, triggers, procedures, and transaction management in solving a real-world social problem.

Importance of Using DBMS over File-Based System

Using a **Database Management System (DBMS)** offers several advantages over traditional file-based systems:

- **Data Integrity:** Ensures accurate and consistent data using constraints and relationships.
- **Reduced Redundancy:** Avoids duplication of data by organizing it into related tables.
- **Efficient Data Retrieval:** Enables fast searching, filtering, and querying using SQL.
- **Security:** Provides controlled access to data, preventing unauthorized usage.
- **Concurrency Control:** Allows multiple users to access the system without conflicts.
- **Automation:** Supports advanced features like triggers and procedures to automate tasks such as case logging, status updates, and rehabilitation tracking..

In contrast, file-based systems lack proper structure, are difficult to maintain, and are prone to data inconsistency and loss. Therefore, a DBMS-based solution is more reliable and suitable for managing such real-world applications.

Problem Statement

Definition of the Real-World Problem

The issue of missing persons, especially children, elderly individuals, and victims of trafficking, is a major social concern. Information related to missing persons is often scattered across multiple police stations, NGOs, shelters, and rehabilitation centres, making coordination difficult and inefficient. Existing systems rely heavily on manual records or isolated databases, which can delay identification, rescue, rehabilitation, and reunification processes. The absence of a centralized system also makes it difficult to maintain proper case histories, track updates, and share information among organizations.

Limitations of the Existing/Manual System

The current manual and disconnected systems have several limitations:

- Lack of centralized storage and coordination
- Difficulty in tracking case progress and updates
- Delayed communication between authorities and NGOs
- High possibility of duplicate or inconsistent records
- Limited transparency and accountability
- Time-consuming search and retrieval process

These issues reduce the efficiency of rescue operations and create challenges in ensuring proper case handling and reunification.

Scope of Database Usage in Solving the Problem

A Database Management System (DBMS) can effectively address these challenges by providing a centralized and structured solution. The proposed system can:

- Store and organize missing person records efficiently
- Maintain relationships between cases, sightings, shelters, and NGOs
- Enable quick searching and filtering of information
- Support audit logging and automated status tracking using triggers
- Improve coordination among police stations, shelters, and NGOs
- Ensure data consistency, security, and integrity

Thus, the use of a DBMS significantly improves the efficiency, reliability, and transparency of managing missing person and reunification cases.

Objectives of the Project

The main objectives of this project are:

- To design a database system using Entity–Relationship (ER) data modeling
- To convert the ER model into relational database tables
- To apply normalization techniques up to Third Normal Form (3NF) to reduce redundancy
- To implement the database using Structured Query Language (SQL)
- To use advanced database features such as procedures, functions, triggers, and cursors
- To ensure data consistency and integrity using constraints and transactions
- To develop a centralized system for managing missing person, shelter, NGO, and sponsorship records

Scope of the Project

The scope of the Missing Person & Reunion Tracking System is limited to managing missing person records, sightings, rehabilitation details, sponsorship information, and reunification processes using a centralized database approach. The system focuses primarily on backend database operations along with a simple web-based interface for interaction.

- **Functional Boundaries of the System:**
The system handles registration of missing person cases, sighting management, rehabilitation tracking, sponsorship management, reunion records, and audit logging. It focuses on maintaining structured and centralized data for efficient coordination among different organizations involved in rescue and rehabilitation activities.
- **Types of Users Involved:**
 - **Police Officers:** Register and update missing person cases, manage investigations, and verify reports
 - **NGO Staff:** Assist in rehabilitation, sponsorship coordination, and case tracking
 - **Shelter Administrators:** Maintain shelter and rehabilitation records
 - **Admin:** Manage users, monitor activities, and maintain overall system operations
 - **Sponsors/Support Partners:** Provide financial or welfare support for rehabilitation programs
- **Modules Covered:**
 - User management
 - Missing person management
 - Sighting management
 - NGO and shelter coordination
 - Rehabilitation tracking
 - Sponsorship management
 - Reunion and case closure management
 - Audit logging and activity tracking
 - Report generation and case status monitoring

Proposed System Description

The proposed system is a database-driven solution designed to manage and monitor missing person cases efficiently through a centralized platform. The system integrates police departments, NGOs, shelters, rehabilitation centres, and sponsors to improve coordination, transparency, and case tracking. It automates important operations related to missing person management, rehabilitation workflows, sponsorship tracking, and reunification processes.

- **Overall Working of the System:**

Missing person cases are registered and stored in a centralized database. Authorities and NGOs can update sightings, rehabilitation details, sponsorship records, and case progress information. The system maintains relationships between persons, cases, shelters, NGOs, and sponsors using relational database concepts. Triggers and procedures automate activities such as case logging, status updates, and reunification tracking. The frontend interface allows users to interact with the database and retrieve information efficiently.

- **Key Features and Functionalities:**

- Centralized database for managing missing person and rehabilitation records
- Efficient searching, filtering, and case tracking using SQL queries
- Management of sightings, shelters, NGOs, sponsorships, and reunion records
- Automated status updates and audit logging using triggers
- Use of procedures, functions, and transactions for better control and automation
- Secure and structured data storage using relational constraints and normalization
- Web-based interface for user interaction and data management

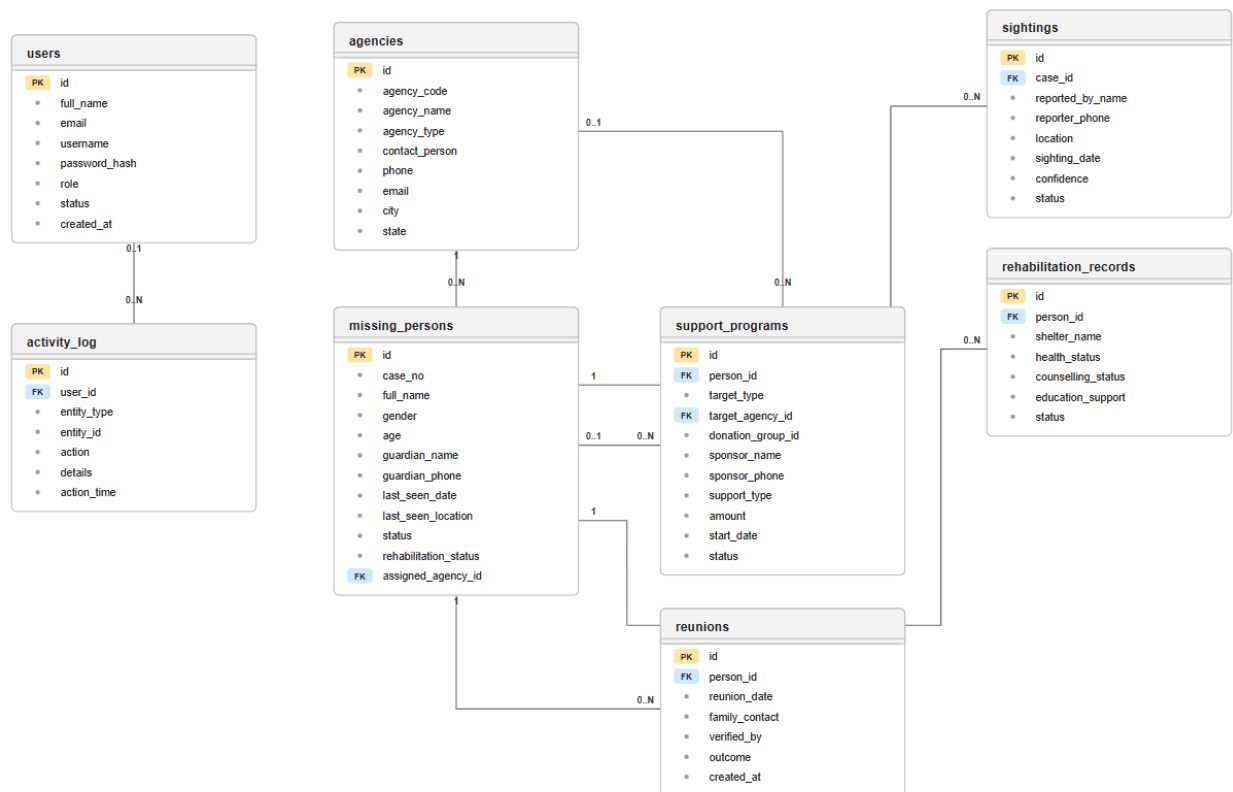
- **Improvement Over Existing System:**

The proposed system improves efficiency by replacing fragmented and manual record-keeping methods with a centralized relational database system. It enhances coordination between police departments, NGOs, and shelters while reducing redundancy and inconsistency through normalization and relational constraints. Automated logging and tracking mechanisms improve transparency, accountability, and reliability in managing missing person and reunification cases.

Entities

The main entities identified in the system are:

- Users
- Agencies
- Missing Persons
- Sightings
- Rehabilitation Records
- Support Programs / Sponsorships
- Reunions
- Activity Logs



1.USERS Table

users	
PK	id
	• full_name
	• email
	• username
	• password
	• password_hash
	• role
	• status
	• created_at

- Converted from the **Users** entity in the ER diagram
- **Primary Key:** id
- Stores login credentials, user roles, and account status information
- Used for authentication and role-based access control within the system
- Related to system activities and operations through one-to-many relationships
- Supports different user categories such as admin, agency staff, and support partners
- Maintains account creation and activity tracking information using timestamps

2.AGENCIES Table

agencies	
PK	id
	• agency_code
	• agency_name
	• agency_type
	• contact_person
	• phone
	• email
	• address
	• city
	• state
	• created_at

- Converted from the **Agencies** entity in the ER diagram
- **Primary Key:** id
- Stores information related to police stations, NGOs, shelters, hospitals, and rehabilitation centres
- Maintains agency details such as contact information, address, city, and state
- Used to coordinate missing person cases, rehabilitation activities, and reunification processes
- Related to the **Missing Persons** table through one-to-many (1:M) relationships
- Supports centralized management and communication among different organizations involved in the system

3. MISSING_PERSONS Table

missing_persons	
PK	id
	• case_no
	• full_name
	• gender
	• age
	• guardian_name
	• guardian_phone
	• last_seen_date
	• last_seen_location
	• identifying_marks
	• photo_url
	• status
	• rehabilitation_status
FK	assigned_agency_id
	• created_at
	• updated_at

- Converted from the **Missing Persons** entity in the ER diagram
- **Primary Key:** `id`
- **Foreign Key:** `assigned_agency_id` → `agencies(id)`
- Stores detailed information related to missing individuals, including personal details, guardian information, last seen location, identifying marks, and rehabilitation status
- Maintains case-related information such as case number, status, assigned agency, and timestamps for tracking updates
- Related to sightings, rehabilitation records, support programs, and reunion records through one-to-many (1:M) relationships
- Supports centralized case management, rehabilitation tracking, and reunification processes within the system

4. SIGHTINGS Table

sightings	
PK	id
FK	case_id
	reported_by_name
	reporter_phone
	location
	sighting_date
	confidence
	status
	notes
	created_at

- Converted from the **Sightings** entity in the ER diagram
- **Primary Key:** `id`
- **Foreign Key:** `case_id` → `missing_persons(id)`
- Stores information related to reported sightings of missing persons, including location, date, reporter details, confidence level, and verification status
- Helps authorities track possible locations and movements of missing individuals during investigations
- Maintains verification and status tracking for each sighting report
- Related to the **Missing Persons** table through a one-to-many (1:M) relationship, where one missing person case can have multiple sighting records
- Supports efficient case monitoring and investigation workflows within the system

5. REHABILITATION_RECORDS Table

rehabilitation_records	
PK	id
FK	person_id
	• shelter_name
	• health_status
	• counselling_status
	• education_support
	• status
	• notes
	• updated_at

- Converted from the **Rehabilitation Records** entity in the ER diagram
- **Primary Key:** id
- **Foreign Key:** person_id → missing_persons(id)
- Stores rehabilitation-related information such as shelter details, health condition, counselling status, education support, and rehabilitation progress
- Maintains recovery and support records for rescued or reunited individuals
- Helps authorities and NGOs monitor rehabilitation activities and welfare support efficiently
- Related to the **Missing Persons** table through a one-to-many (1:M) relationship
- Supports rehabilitation tracking, follow-up management, and recovery process monitoring within the system

6. SUPPORT_PROGRAMS Table

support_programs	
PK	id
FK	person_id
	• target_type
FK	target_agency_id
	• donation_group_id
	• sponsor_name
	• sponsor_phone
	• support_type
	• amount
	• start_date
	• status
	• notes

- Converted from the **Support Programs** entity in the ER diagram
- **Primary Key:** id
- **Foreign Keys:**
 - person_id → missing_persons(id)
 - target_agency_id → agencies(id)
- Stores sponsorship and welfare support details provided to missing or rehabilitated individuals and organizations
- Maintains information related to sponsors, support type, donation amount, support duration, and program status

- Supports financial aid, educational support, medical assistance, shelter aid, and rehabilitation programs
- Related to both **Missing Persons** and **Agencies** tables through one-to-many (1:M) relationships
- Helps manage rehabilitation support and welfare coordination within the system

7. REUNIONS Table

reunions	
PK	id
FK	person_id
	• reunion_date
	• family_contact
	• verified_by
	• outcome
	• notes
	• created_at

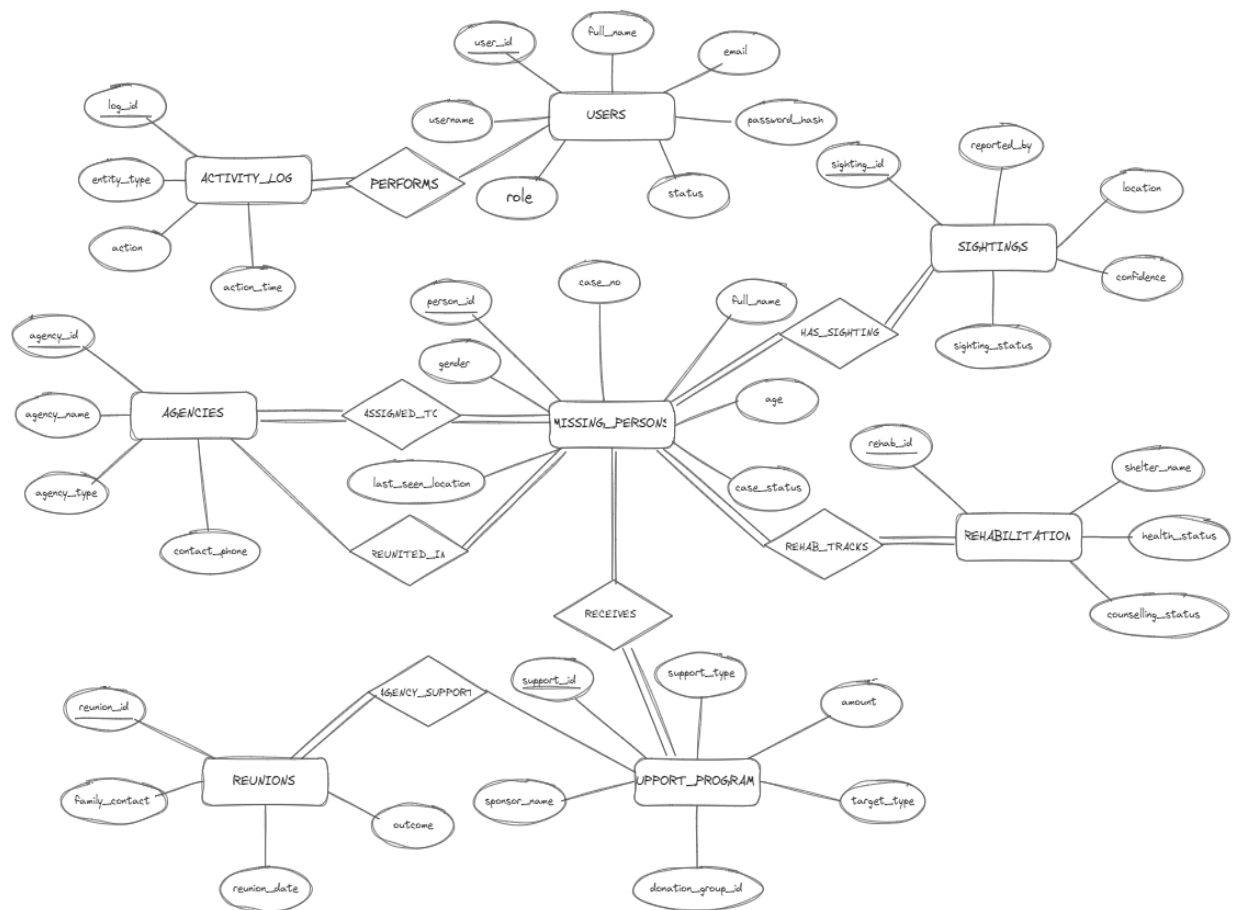
- Converted from the **Reunions** entity in the ER diagram
- **Primary Key:** id
- **Foreign Key:** person_id → missing_persons(id)
- Stores information related to reunification and case closure processes
- Maintains details such as reunion date, family contact information, verification details, outcome status, and additional notes
- Helps authorities and NGOs track successful reunifications and follow-up activities
- Related to the **Missing Persons** table through a one-to-many (1:M) relationship
- Supports case completion tracking, verification management, and reunification monitoring within the system

8. ACTIVITY_LOG Table

activity_log	
PK	id
FK	user_id
	• entity_type
	• entity_id
	• action
	• details
	• action_time

- Converted from the **Activity Log** entity in the ER diagram
- **Primary Key:** id
- **Foreign Key:** user_id → users(id)
- Stores audit and activity tracking information related to system operations
- Maintains details such as performed action, related entity type, entity identifier, action description, and timestamp
- Helps monitor user activities and maintain transparency within the system
- Supports accountability and audit logging for important database operations and case updates
- Related to the **Users** table through a one-to-many (1:M) relationship, where one user can perform multiple activities within the system

E-R DIAGRAM :



PL/SQL Components

Stored Procedures

Definition:

A stored procedure is a precompiled set of SQL statements used to perform a specific task within the database system.

```
CREATE PROCEDURE mark_case_reunited(  
    IN p_person_id INT,  
    IN p_family_contact VARCHAR(160),  
    IN p_verified_by VARCHAR(140),  
    IN p_notes TEXT  
)  
BEGIN  
    INSERT INTO reunions(  
        person_id,  
        reunion_date,  
        family_contact,  
        verified_by,  
        outcome,  
        notes  
    )  
    VALUES (  
        p_person_id,  
        CURDATE(),  
        p_family_contact,  
        p_verified_by,  
        'Reunited',  
        p_notes  
    );  
END
```

Explanation:

The `mark_case_reunited` procedure is used to insert reunion details into the `reunions` table whenever a missing person is successfully reunited with their family. The procedure records important information such as the person ID, family contact details, verification information, reunion outcome, and additional notes. It also automatically stores the current date as the reunion date using the `CURDATE()` function.

This procedure helps automate the reunification process, reduces manual data entry, and ensures consistency in maintaining reunion records within the system.

Triggers

Definition:

A trigger is a database object that automatically executes when a specified event such as INSERT, UPDATE, or DELETE occurs on a table.

```
CREATE TRIGGER trg_sighting_verified
AFTER UPDATE ON sightings
FOR EACH ROW
> BEGIN
>   IF NEW.status = 'Verified' THEN
      UPDATE missing_persons
      SET status = 'Sighted'
      WHERE id = NEW.case_id
      AND status = 'Missing';
- END IF;
- END
```

Explanation:

The `trg_sighting_verified` trigger is executed automatically whenever a sighting record is updated in the `sightings` table. If the status of a sighting is changed to "Verified", the trigger automatically updates the corresponding record in the `missing_persons` table and changes the person's status from "Missing" to "Sighted".

This trigger helps automate case tracking and reduces the need for manual status updates. It improves consistency, ensures faster synchronization between related tables, and enhances the efficiency of the missing person investigation process within the system.

```
CREATE TRIGGER trg_reunion_insert
AFTER INSERT ON reunions
FOR EACH ROW
BEGIN
    IF NEW.outcome = 'Reunited' THEN
        UPDATE missing_persons
        SET status = 'Reunited',
            rehabilitation_status = 'In Progress'
        WHERE id = NEW.person_id;
    END IF;
END
```

Explanation:

The `trg_reunion_insert` trigger is executed automatically after a new record is inserted into the `reunions` table. If the reunion outcome is marked as "Reunited", the trigger automatically updates the corresponding record in the `missing_persons` table by changing the status to "Reunited" and updating the rehabilitation status to "In Progress".

This trigger helps automate the reunification workflow and ensures that related records remain synchronized across tables. It reduces manual updates, improves consistency, and supports efficient case closure and rehabilitation tracking within the system.

Tools & Technologies Used

The project is implemented using **MySQL** as the primary Database Management System (DBMS) due to its reliability, ease of use, and wide adoption. SQL is used for performing all database operations including data definition (DDL), data manipulation (DML), and data retrieval through queries. Advanced SQL features such as **joins, aggregate functions, views, triggers, stored procedures, and cursors** are also utilized to enhance functionality and automation.

For development and execution, **MySQL Workbench** is used as the interface tool. It provides a user-friendly environment for designing schemas, writing queries, visualizing ER diagrams, and managing the database efficiently.

In addition to the database implementation, a web-based frontend is developed using **HTML, CSS, and JavaScript** to provide user interaction with the system. The backend is built using **Node.js** and **Express.js**, which connects the frontend with the MySQL database and handles application logic such as login, role-based access, case registration, sightings, rehabilitation records, support programs, and reunion tracking. The application runs through an Express server, where frontend pages and API routes are served together for smooth communication between the user interface and the database.

Expected Outcomes

The expected outcome of this project is the successful development of a well-structured and **centralized database system** for managing **missing person records, rehabilitation details, sponsorship information, and reunification processes**. The system is designed to ensure efficient storage, organization, and retrieval of data using optimized SQL queries and relational database concepts. By integrating information from **police departments, NGOs, shelters, and rehabilitation centres** into a single platform, the project aims to improve coordination and transparency among different organizations involved in rescue and recovery operations.

Automation features such as **triggers, stored procedures, functions, and audit logging** help reduce manual effort by automatically updating case statuses, maintaining activity records, and supporting rehabilitation and reunification workflows. The use of **primary keys, foreign keys, normalization, constraints, and transaction management** ensures data consistency, integrity, reliability, and security throughout the system. These features help minimize redundancy and improve the accuracy of case-related information.

The project also aims to provide a user-friendly web-based interface for efficient interaction with the database system. Through the integration of frontend and backend technologies, users such as police officers, NGO staff, shelter administrators, and system administrators can register cases, update records, monitor rehabilitation progress, manage support programs, and track reunification activities in a structured manner. Overall, the system is expected to provide a scalable and practical solution for handling missing person cases while demonstrating the real-world application of DBMS concepts, automation techniques, and modern web technologies.